

Lecture 03

Systems of Linear Equations I: Forms and Elimination

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Recall

In the last lectures ...

- Definition of norm and dot (inner product)
- Calculation of vector norms and inner products
- Real-world examples

Today

Today ... System of **Linear Equations!** (& a bit inner product)

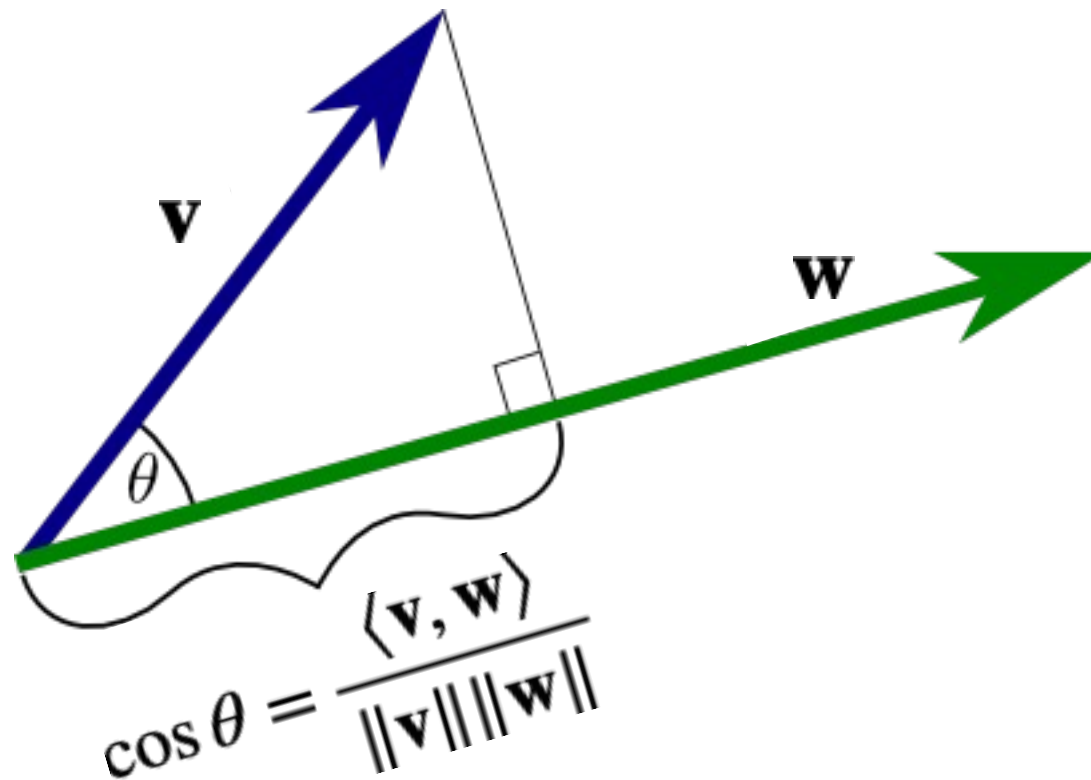
After this lecture, you should be able to

1. Write the 4 forms of **systems of linear equations**
2. Write various forms of **matrix-vector product**

Part 0 Inner Product and Norm

Cosine Similarity

Cosine Similarity $\longrightarrow \cos \theta = \frac{\langle \mathbf{v}, \mathbf{w} \rangle}{\|\mathbf{v}\| \|\mathbf{w}\|}$



Range: $[-1, 1]$. The higher, the more similar.

Example

Given three vectors, which two pairs are more similar?

A. q and u

B. q and v

$$(0.8, 1.1, 0.5) = \mathbf{q}$$

$$(2.5, 4.3, 3.7) = \mathbf{u}$$

$$(-1.5, 2.5, -0.3) = \mathbf{v}$$

Example

Given three vectors, which two pairs are more similar?

A. q and u

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$$(0.8, 1.1, 0.5) = \mathbf{q}$$

$$(2.5, 4.3, 3.7) = \mathbf{u}$$

$$(-1.5, 2.5, -0.3) = \mathbf{v}$$

Cosine similarity of q and u is ~ 0.95 ;

Cosine similarity of q and v is ~ 0.33 .

Compare cosine similarity: $0.95 > 0.33$.

so candidate 1 is more similar to the query.

Calculation of Cosine Similarity

Just list the procedure for the first pair here

First Pair: (0.8, 1.1, 0.5) and (2.5, 4.3, 3.7)

1. **Dot Product:**

$$\mathbf{A} \cdot \mathbf{B} = 0.8 \times 2.5 + 1.1 \times 4.3 + 0.5 \times 3.7 = 2 + 4.73 + 1.85 = 8.58$$

1. **Euclidean Norms:**

$$\|\mathbf{A}\| = \sqrt{0.8^2 + 1.1^2 + 0.5^2} = \sqrt{0.64 + 1.21 + 0.25} = \sqrt{2.1} \approx 1.4491$$

$$\|\mathbf{B}\| = \sqrt{2.5^2 + 4.3^2 + 3.7^2} = \sqrt{6.25 + 18.49 + 13.69} = \sqrt{38.43} \approx 6.2006$$

1. **Cosine Similarity:**

$$\text{Cosine Similarity} = \frac{8.58}{1.4491 \times 6.2006} \approx \frac{8.58}{8.9848} \approx 0.9549$$

Property 1: Relation of Dot Product and Norm

Cauchy–Schwarz Inequality

$$|\langle \mathbf{v} \cdot \mathbf{w} \rangle| \leq \|\mathbf{v}\| \|\mathbf{w}\|$$

An “intuitive” but not rigorous explanation:

$$\cos \theta = \frac{\langle \mathbf{v} \cdot \mathbf{w} \rangle}{\|\mathbf{v}\| \|\mathbf{w}\|} \leq 1$$

We will formally prove Cauchy–Schwarz inequality in a future lecture.
Accept it for now.

Property 2: Triangle Inequality

Triangle inequality

$$\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\|$$

Proof: [show proof in class]

Recall. $\langle \vec{u}, \vec{u} \rangle = \|\vec{u}\|^2$ ①

$$\begin{aligned} \|\vec{v} + \vec{w}\|^2 &\stackrel{\text{①}}{=} \langle \vec{v} + \vec{w}, \vec{v} + \vec{w} \rangle \\ &= \langle \vec{v} + \vec{w}, \vec{v} \rangle + \langle \vec{v} + \vec{w}, \vec{w} \rangle \quad [\text{Linearity}] \\ &= \langle \vec{v}, \vec{v} \rangle + \langle \vec{w}, \vec{v} \rangle + \langle \vec{v}, \vec{w} \rangle + \langle \vec{w}, \vec{w} \rangle \quad [\text{Linearity}] \\ &= \|\vec{v}\|^2 + 2\langle \vec{w}, \vec{v} \rangle + \|\vec{w}\|^2 \quad [\text{by ① \& symmetry}] \\ &\leq \|\vec{v}\|^2 + 2\|\vec{w}\|\|\vec{v}\| + \|\vec{w}\|^2 \quad [\text{C-S ineq.}] \\ &= \|\vec{v} + \vec{w}\|^2. \quad \square \end{aligned}$$

Property 3: Pythagoras Law (毕达哥拉斯定理)

Pythagoras Law

$$\|v\|^2 + \|w\|^2 = \|v - w\|^2 \quad \text{iff} \quad \langle \mathbf{v}, \mathbf{w} \rangle = 0$$

Proof [[skip the proof in class](#)]

Part I System of Linear Equations

Linear System of Equations: Preliminary School Example

Problem (Chicken-Rabbit Problem 鸡兔同笼)

There are 16 heads and 36 feet in a cage.
How many chickens and how many rabbits are there?
Write the equations and solve it.

Assumption: Each chicken has 1 head and 2 feet, and each rabbit has 1 head and 4 feet.

Suppose # chicken is x_1 , # rabbits is x_2 .

$$\begin{cases} x_1 + x_2 = 16 & \textcircled{1} \\ 2x_1 + 4x_2 = 36 & \textcircled{2} \end{cases}$$

$$\textcircled{2} - 2 \times \textcircled{1}, \text{ get } 2x_2 = 4 \Rightarrow x_2 = 2. \quad \textcircled{3}$$

$$\begin{aligned} \textcircled{1} - \textcircled{3}, \text{ get } & x_1 = 14. \\ \text{Solution.} & \begin{cases} x_1 = 14 \\ x_2 = 2 \end{cases} \end{aligned}$$

Linear Equations

Definition (Linear Equations)

A **linear equation** is the equation of the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where $a_1, a_2, \dots, a_n, b$ are real numbers and $x_1, x_2, \dots, x_n$ are variables

Write it in the vector form:

$$\langle \vec{a}, \vec{x} \rangle = b, \text{ where}$$
$$\vec{a} = (a_1, \dots, a_n)$$
$$\vec{x} = (x_1, \dots, x_n)$$

Exercise

Exercise (Linear Equations)

Are the following linear equations?

1. $-x_1 + 4x_4 = 2x_2 + 3x_3$ ✓

x_1, x_2, x_3, x_4 are variables



Exercise

Exercise (Linear Equations)

Are the following linear equations?

1. $-x_1 + 4x_4 = 2x_2 + 3x_3$

2. $-x_1x_4 = 2x_2 + 3x_3$



x_1, x_2, x_3, x_4 are variables

x_1, x_2, x_3, x_4 are variables

Exercise

Exercise (Linear Equations)

Are the following linear equations?



1. $-x_1 + 4x_4 = 2x_2 + 3x_3$

x_1, x_2, x_3, x_4 are variables

2. $-x_1x_4 = 2x_2 + 3x_3$

x_1, x_2, x_3, x_4 are variables

3. $a_1x_1 + a_2x_2 = b$



x_1, x_2, a_1, a_2, b are variables

System of Linear Equations

Definition (System of Linear Equations)

An $m \times n$ **system of linear equations** is a *collection* of m linear equations with n variables

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$

...

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m$$

where all a_{ij} and b_i are real numbers and $x_1, x_2, \dots, x_n$ are variables

(A system of linear equations can be called a **linear system** for short)

Exercise

Q1 In a 8×12 system of linear equations, there are ____ variables?

A. 8

B. 12

C. 96

D. None of the above

Exercise

Q1 In a 8×12 system of linear equations, there are ____ variables?

- A. 8 B. 12 C. 96 D. None of the above

Q2 Is the following a linear system?

$$-a_{11} + 4a_{12} = 2a_{13} + 3a_{14}$$

$$5a_{11} + 3a_{12} = a_{13} + 3a_{14}$$

where $a_{11}, a_{12}, a_{13}, a_{14}$ are variables

yes



Exercise

Q1 In a 8×12 system of linear equations, there are ____ variables?

- A. 8 B. 12 C. 96 D. None of the above

Q2 Is the following a linear system?

$$-a_{11} + 4a_{12} = 2a_{13} + 3a_{14}$$

$$5a_{11} + 3a_{12} = a_{13} + 3a_{14}$$

where $a_{11}, a_{12}, a_{13}, a_{14}$ are variables



It is critical to know what are the **variables (unknowns)**!

What is New in this Course?

You learned these in middle school (or even primary school).

What more can we study? What will be new?

1. Practice.

You can solve system of equations in 2 variables.

What about 5 variables? What about 100 variables?

What is a general method to solve an any-variable system? (For computers)

2. Theory.

Does your general method always work? Can you prove it?

First step: To answer these questions, we will rewrite system of equations with vectors and matrices.

Row-Vector Form of System

(F1) scalar form

$$x_1 + x_2 = 14$$

$$2x_1 + 4x_2 = 36$$

row-reduction $\rightarrow$

(F2) Row vector form

$$[1, 1] \bullet [x_1, x_2] = 14$$

$$[2, 4] \bullet [x_1, x_2] = 36$$

dot product

Row-Vector Form of System

(F1) scalar form

$$\begin{aligned}x_1 + x_2 &= 14 \\ 2x_1 + 4x_2 &= 36\end{aligned}$$

row-reduction $\longrightarrow$

(F2) Row vector form

•
•
dot product

Can we make the form even simpler?

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}, \quad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} 14 \\ 36 \end{bmatrix}$$

Define $A\vec{x} \triangleq \begin{bmatrix} x_1 + x_2 \\ 2x_1 + 4x_2 \end{bmatrix}$, then the system becomes

(F0) matrix form:

$$A\vec{x} = \vec{b}$$

Column-Vector Form

$$x_1 + x_2 = 14$$

$$2x_1 + 4x_2 = 36$$

Another way of writing the equations: Column-vector form

Let $a_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $a_2 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$. Rewrite the system. [in-class exercise]

(F3) Column vector form

$$x_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 14 \\ 36 \end{bmatrix}, \text{ or } x_1 \vec{a}_1 + x_2 \vec{a}_2 = \vec{b}$$

LHS (Left-Hand Side) is Linear combination of vectors

Four Forms of Linear Systems

Scalar form

$$\begin{cases} x_1 + x_2 = 16, \\ 2x_1 + 4x_2 = 36. \end{cases}$$

Row-vector form

(Unknown vector satisfies n linear equations simultaneously)

$$\begin{cases} \langle [1, 1], [x_1, x_2] \rangle = 16 \\ \langle [2, 4], [x_1, x_2] \rangle = 36 \end{cases}$$

Column-vector form

(Unknown combination of columns produces vector b)

$$x_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 16 \\ 36 \end{bmatrix}$$

Matrix form

(Given matrix times unknown vector produces b)

$$A \vec{x} = b.$$

[details next]

Part II Matrix-Vector Product & Four Forms of Linear Systems

Textbook v5: Sec. 1.3 (only first half) and Sec. 2.1

Matrix Definition

Definition 4.1 (Matrix)

An $m \times n$ **matrix** A is a rectangular array of **numbers** with m rows and n columns in the following form:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} =: (a_{ij})_{m \times n} \in \mathbb{R}^{m \times n}.$$

where all a_{ij} are scalars.

Def 4.2 (Dimension of matrix & vector):

For an $m \times n$ **matrix**, the dimension of A is $m \times n$ (reads “m by n”).

For an $m \times 1$ vector, the dimension of the vector is m , or $m \times 1$.

Matrix: Example and non-example

Often use square bracket

$$\begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0.5 & -3.7 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -2.3 & 0.1 \\ 1.3 & 4 & -0.1 & 0 \\ 4.1 & -1 & 0 & 1.7 \end{bmatrix}$$

Can also use round bracket

$$\begin{pmatrix} 1 & 1 \\ 2 & 4 \end{pmatrix}$$

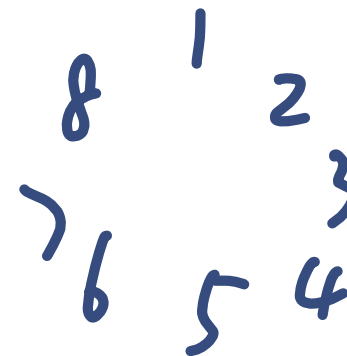
$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0.5 & -3.7 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 & -2.3 & 0.1 \\ 1.3 & 4 & -0.1 & 0 \\ 4.1 & -1 & 0 & 1.7 \end{pmatrix}$$

Non-examples:



Yang-hui triangle



round-table

Matrix Notation

Matrices are often denoted by $A, B, C, \dots$

For the matrix $A = (a_{ij})_{1 \leq i \leq m, 1 \leq j \leq n}$, a_{ij} is called the (i, j) -th entry (element) of A .

When $m = n$, A is called a *square matrix*; a *rectangular matrix* o.w.

When all entries are zeros, A is called a *zero matrix*.

Matrix v.s. Vector v.s. Scalar

- When $m = 1$, A is a **row vector**
- When $n = 1$, A is a **column vector**

Matrix: general $m \times n$

Vector: $m \times 1$ or $1 \times n$ matrix

Scalar: 1×1 matrix

Remark: In python, scalar and 1×1 matrix are different!

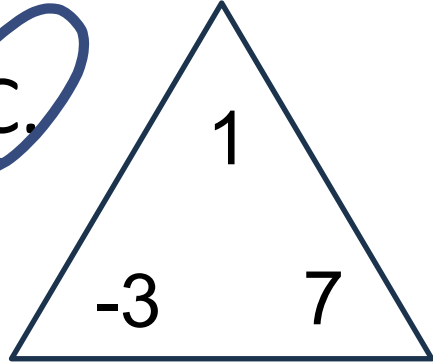
e.g. scalar **3.5** v.s. 1×1 matrix **[[3.5]]**

(Easily cause bug if you don't know this!)

Exercise

Q1 The dimension of $A = \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}$ is 2×2
A. 4 B. 2 C. 1 **D. None of the above**

Q2 Suppose $m, n > 5$.
In a matrix $A = (a_{ij})_{m \times n}$, the entry $a_{3,5}$ is in 5'th column.
A. 3 **B. 5** C. 15 D. none of the above

Q3 Which of the following is NOT a matrix?
A. $\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$ B. $[3 \ 5 \ 7]$ **C.**  D. $[-3.5]$

Matrix Conventions

- Column of a matrix $a_j = \begin{bmatrix} A_{1j} \\ \vdots \\ A_{mj} \end{bmatrix}$
- Row of a matrix $a^{(i)} = \begin{bmatrix} A_{i1} & \cdots & A_{in} \end{bmatrix}$

The skill to identify matrices' columns and matrices' rows is important for the future, but often ignored!

Matrix v.s. Column Vectors: Example of 2 by 3 Matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$$a_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \quad a_2 = \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \quad a_3 = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

Express A with its columns. (in-class exercise)

Answer: $A = [\vec{a}_1, \vec{a}_2, \vec{a}_3]$

Imperfect answer: $A = [\vec{a}_1 \vec{a}_2 \vec{a}_3]$
(comma missing)

Matrix v.s. Row Vectors: Example of 2 by 3 Matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$$a^{(1)} = [1 \quad 2 \quad 3], \quad a^{(2)} = [4 \quad 5 \quad 6]$$

Then $A =$ $\begin{bmatrix} \vec{a}^{(1)} \\ \vec{a}^{(2)} \\ \vec{a}^{(3)} \end{bmatrix}$

Definition of Matrix-Vector Product

Definition 3.1 (Matrix-vector product definition)

Let $u = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}$ and A be an $m \times n$ matrix $A = (a_{ij})$, where $i = 1, \dots, m$; $j = 1, \dots, n$,

The matrix-vector product Au is an $m \times 1$ column vector:

$$Au = \begin{bmatrix} \sum_{j=1}^n a_{1j}u_j \\ \sum_{j=1}^n a_{2j}u_j \\ \vdots \\ \sum_{j=1}^n a_{mj}u_j \end{bmatrix}.$$

Remark: Dimension checking

$$\begin{matrix} A & u & = & v \\ m \times n & n \times 1 & & m \times 1 \end{matrix}$$

Example:

$$\begin{bmatrix} 1 & 2 & 1 \\ 3 & 4 & 3 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \times 5 + 2 \times (-1) + 1 \times 6 \\ 3 \times 5 + 4 \times (-1) + 3 \times 6 \end{bmatrix} = \begin{bmatrix} 9 \\ 29 \end{bmatrix}$$

Exercise

Q1 For which of the following vector $\mathbf{v}$, the product $\mathbf{Z}\mathbf{v}$ is valid? Here $\mathbf{Z} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

A. $\mathbf{v} = \begin{bmatrix} 1 \\ -5 \\ 3 \end{bmatrix}$

B. $\mathbf{v} = [1, 5, 7]$

C. $\mathbf{v} = [1, 3]$

D. None of the above

Exercise

Q1 For which of the following vector $\mathbf{v}$, the product $\mathbf{Z}\mathbf{v}$ is valid? Here $\mathbf{Z} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

A. $\mathbf{v} = \begin{bmatrix} 1 \\ -5 \\ 3 \end{bmatrix}$

B. $\mathbf{v} = [1, 5, 7]$

C. $\mathbf{v} = [1, 3]$

D. None of the above

Q2 For which of the following matrix M , the product $M\mathbf{v}$ is valid? Here $\mathbf{v} = \begin{bmatrix} 1 \\ -5 \end{bmatrix}$.

A. $M = \begin{bmatrix} 1 & 2 & 5 \\ 3 & 4 & 2 \end{bmatrix}$

B. $M = \begin{bmatrix} 1 & 5 & 4 \\ 3 & -3.5 & -4 \\ 3 & 2 & 8 \end{bmatrix}$

C. $M = [7, 3]$

D. None of the above

Matrix-Vector Product and Linear System

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad b = \begin{bmatrix} 14 \\ 36 \end{bmatrix}$$

$$\begin{cases} x_1 + x_2 = 16, \\ 2x_1 + 4x_2 = 36. \end{cases}$$

$$\longleftrightarrow Ax = b$$

where $Ax = \begin{bmatrix} x_1 + x_2 \\ 2x_1 + 4x_2 \end{bmatrix}$

Matrix-Vector Product and Linear System

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad b = \begin{bmatrix} 14 \\ 36 \end{bmatrix}$$

$$\begin{cases} x_1 + x_2 = 16, \\ 2x_1 + 4x_2 = 36. \end{cases}$$

$$\longleftrightarrow Ax = b$$

where $Ax =$

$$\begin{cases} [1, 1] \cdot [x_1, x_2] = 16, \\ [2, 4] \cdot [x_1, x_2] = 36. \end{cases}$$

$$\longleftrightarrow Ax = b$$

where $Ax =$

$$\begin{bmatrix} \langle [1, 1], [x_1, x_2] \rangle \\ \langle [2, 4], [x_1, x_2] \rangle \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} x_1 + \begin{bmatrix} 1 \\ 4 \end{bmatrix} x_2 = \begin{bmatrix} 16 \\ 36 \end{bmatrix}.$$

$$\longleftrightarrow Ax = b$$

where $Ax =$

$$x_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

Row-form of Matrix-Vector Product

Property 1 (Row-form of Matrix-vector product)

Let A be an $m \times n$ matrix and $\mathbf{w}$ be an $n \times 1$ column vector.

Suppose the matrix A can be represented as

$$A = \begin{bmatrix} \mathbf{a}^{(1)} \\ \mathbf{a}^{(2)} \\ \vdots \\ \mathbf{a}^{(m)} \end{bmatrix}. \quad \text{Then} \quad A\mathbf{w} = \begin{bmatrix} \mathbf{a}^{(1)}\mathbf{w} \\ \mathbf{a}^{(2)}\mathbf{w} \\ \vdots \\ \mathbf{a}^{(m)}\mathbf{w} \end{bmatrix}$$

Property 2 (Column-form of Matrix-vector product)

Let A be an $m \times n$ matrix and $\mathbf{w}$ be an $n \times 1$ column vector.

Suppose A is represented as: $A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \dots \ \mathbf{a}_n]$,

then $A\mathbf{w} = w_1\mathbf{a}_1 + w_2\mathbf{a}_2 + \dots + w_n\mathbf{a}_n$.

Example

$$A = \begin{bmatrix} 1 & 4 & 2 & 3 & 5 \\ -2 & 1 & 3 & 0 & -1 \\ 0 & 7 & -1 & -2 & 4 \end{bmatrix}, \mathbf{u} = \begin{bmatrix} 1 \\ -2 \\ 0 \\ 5 \\ -1 \end{bmatrix},$$

$$A\mathbf{u} = \begin{bmatrix} 1 * 1 + 4 * (-2) + 2 * 0 + 3 * 5 + 5 * (-1) \\ (-2) * 1 + 1 * (-2) + 3 * 0 + 0 * 5 + (-1) * (-1) \\ 0 * 1 + 7 * (-2) + (-1) * 0 + (-2) * 5 + 4 * (-1) \end{bmatrix}$$

$$A\mathbf{u} = \begin{bmatrix} 3 \\ -3 \\ -28 \end{bmatrix}$$

$$\begin{aligned} \text{Or: } A\mathbf{u} &= 1 \cdot \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} + (-2) \cdot \begin{bmatrix} 4 \\ 1 \\ 7 \end{bmatrix} + 0 \cdot \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} + 5 \cdot \begin{bmatrix} 3 \\ 0 \\ -2 \end{bmatrix} + (-1) \cdot \begin{bmatrix} 5 \\ -1 \\ 4 \end{bmatrix} \\ &= \begin{bmatrix} 3 \\ -3 \\ -28 \end{bmatrix} \end{aligned}$$

Part II-Extra

Dot Product and Matrix- Vector Product

Exercise

Judgement: True or False.

Q1 Let A be an $m \times n$ matrix and $\mathbf{w}$ be an $n \times 1$ column vector.
Suppose A is represented as: $A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \dots \ \mathbf{a}_n]$,
then $A\mathbf{w} = w_1\mathbf{a}_1 + w_2\mathbf{a}_2 + \dots + w_n\mathbf{a}_n$.

T

Q2 Let A be an $m \times n$ matrix and $\mathbf{w}$ be an $n \times 1$ column vector.
Suppose the matrix A can be represented as

$$A = \begin{bmatrix} \mathbf{a}^{(1)} \\ \mathbf{a}^{(2)} \\ \vdots \\ \mathbf{a}^{(m)} \end{bmatrix}. \quad \text{Then} \quad A\mathbf{w} = \begin{bmatrix} \mathbf{a}^{(1)} \cdot \mathbf{w} \\ \mathbf{a}^{(2)} \cdot \mathbf{w} \\ \vdots \\ \mathbf{a}^{(m)} \cdot \mathbf{w} \end{bmatrix}$$

F

Analysis of Exercise

Q2 Let A be an $m \times n$ matrix and $\mathbf{w}$ be an $n \times 1$ column vector.

Suppose the matrix A can be represented as

$$A = \begin{bmatrix} \mathbf{a}^{(1)} \\ \mathbf{a}^{(2)} \\ \vdots \\ \mathbf{a}^{(m)} \end{bmatrix}. \quad \text{Then} \quad A\mathbf{w} = \begin{bmatrix} \mathbf{a}^{(1)} \cdot \mathbf{w} \\ \mathbf{a}^{(2)} \cdot \mathbf{w} \\ \vdots \\ \mathbf{a}^{(m)} \cdot \mathbf{w} \end{bmatrix}$$

Hint: Check the size of $\underbrace{\mathbf{a}^{(1)}}_{1 \times n}$ and $\underbrace{\mathbf{w}}_{n \times 1}$.

Dot product of row vector and column vector: not defined.

Row Vector * Column Vector?

$$a = \begin{bmatrix} a_1 & a_2 \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Can we multiply them?

Tentative answer 1: Yes, by Definition of inner product: $a \cdot x$.

Row Vector * Column Vector?

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Can we multiply them?

Tentative answer 1: Yes, by Definition of inner product: $a \cdot x$.

This answer is problematic:

inner product of row and column vectors are **NOT defined**.

Row Vector * Column Vector?

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Can we multiply them?

Tentative answer 1: Yes, by Definition of inner product: $a \cdot x$.

This answer is problematic:

inner product of row and column vectors are **NOT defined**.

Tentative answer 2: a is a vector, but also a 1x2 matrix.

view ax as matrix-vector product.

Column vector * column vector?

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

Inner product: $\langle x, y \rangle = x_1 y_1 + x_2 y_2$

Matrix-vector product of x and y :

the matrix-vector product xy is invalid.

The product of two column vectors is **NOT** mentioned in the definition of matrix-vector product.

Valid and Invalid Product

	x	y	Product expression	Product type	Reason of valid or invalid
Valid	$x = [x_1, x_2]$	$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$	$x y$	Matrix-vector product	1×2 matrix times 2×1 vector
invalid	$x = [x_1, x_2]$	$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$	$x \cdot y$	dot product	<i>Dot product of</i> 1×2 vector 2×1 vector not defined
Invalid	$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$	$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$	$x y$	Matrix-vector product	" 2×1 matrix times 2×1 vector" is not defined
valid	$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$	$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$	$x \cdot y$	dot product	dot product of two 2×1 vector vectors is valid

Reading: GPT may mislead you if you are not expert

GPT's definition:

Definition 6.1 (Matrix-vector product Def-1)

A is $m \times n$ matrix, $\vec{b}$ is $n \times 1$ vector,
Suppose $A = \begin{bmatrix} \vec{a}^{(1)} \\ \vdots \\ \vec{a}^{(m)} \end{bmatrix}$, $A \cdot \vec{b} \triangleq \begin{bmatrix} \vec{a}^{(1)} \vec{b} \\ \vdots \\ \vec{a}^{(m)} \vec{b} \end{bmatrix} \in \mathbb{R}^{m \times 1}$

transform to formal writing

$$A\mathbf{b} = \begin{bmatrix} \mathbf{a}^{(1)}\mathbf{b} \\ \mathbf{a}^{(2)}\mathbf{b} \\ \vdots \\ \mathbf{a}^{(m)}\mathbf{b} \end{bmatrix} \in \mathbb{R}^{m \times 1},$$

where each element of the resulting vector $A\mathbf{b}$ is obtained by taking the dot product of the corresponding row vector $\mathbf{a}^{(i)}$ of A with the vector $\mathbf{b}$.

GPT4 adds "dot product" in definition though I did not ask it to do so.

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After an expert points out the issue:

is this really a dot product? Here $\mathbf{a}^{(i)}$ is a row vector!



You are absolutely correct to point out the discrepancy: $\mathbf{a}^{(i)}$ is indeed a **row vector**, and the operation between $\mathbf{a}^{(i)}$ (a row vector) and $\mathbf{b}$ (a column vector) is **matrix-vector multiplication**, not the traditional dot product of two vectors in $\mathbb{R}^n$.

The correct description of the operation is that the product $\mathbf{a}^{(i)}\mathbf{b}$ involves a matrix multiplication between a $1 \times n$ row vector $\mathbf{a}^{(i)}$ and an $n \times 1$ column vector $\mathbf{b}$, yielding a scalar. This operation is sometimes loosely referred to as a dot product, but it's more accurately a matrix-vector multiplication.

Here is a revised and more accurate version of the definition:

This operation is sometimes loosely referred to as a dot product, but it's more accurately a matrix-vector multiplication.

Summary Today

Today, we have learned:

- Cosine similarity; Cauch-Schwartz inequality; triangular inequality and its proof.
- Formulation of systems of linear equations
- Matrix-vector product and four forms of linear system